Date. /

2a)

$$A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 2 \\ 1 & 4 & 6 \\ 6 & 1 & 5 \end{bmatrix}$$

Determinan

$$\det(A) = 3\begin{vmatrix} 4 & 6 \\ 1 & 5 \end{vmatrix} - 6\begin{vmatrix} 1 & 6 \\ 6 & 5 \end{vmatrix} + 2\begin{vmatrix} 1 & 4 \\ 6 & 1 \end{vmatrix}$$

Hitung:

$$= 3(20-6) - 6(5-36) + 2(1-24)$$
$$= 3(14) - 6(-31) + 2(-23)$$
$$= 42 + 186 - 46$$
$$\det(A) = 182$$

2(b)

$$M = \begin{bmatrix} 14 & -31 & -23 \\ 26 & 3 & -33 \\ 26 & 16 & 6 \end{bmatrix}$$

2(c) $$C = \begin{bmatrix} 14 & 31 & -23 \\ -26 & 3 & 33 \\ 26 & -16 & 6 \end{bmatrix}$$

Adjoin (transpose)

$$Adj(A) = \begin{bmatrix} 14 & -26 & 26 \\ 31 & 3 & -16 \\ -23 & 33 & 6 \end{bmatrix}$$

Invers:

$$A^{-1} = \frac{1}{182}\begin{bmatrix} 14 & -26 & 26 \\ 31 & 3 & -16 \\ -23 & 33 & 6 \end{bmatrix}$$

deli得力

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2(e)

$N = 26$

$$P = \begin{bmatrix} 120 & 76 & 200 \\ 80 & 4 & 36 \\ 56 & 90 & 170 \end{bmatrix}$$

enkripsi $E = A \cdot P$

Hasil

$$E = \begin{bmatrix} 952 & 434 & 1146 \\ 776 & 632 & 1364 \\ 1000 & 910 & 2066 \end{bmatrix}$$

$P' = A^{-1} \cdot E$

2(f) <u>matlab</u>

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